

RAKSHITH KUMAR K.N

Senior AI / ML Engineer | Generative AI | Production ML Systems | LLMOps | Multi-Cloud AI | Team Leadership
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PROFESSIONAL SUMMARY

Senior AI/ML Engineer with **5+ years** delivering **\$2.3M+ measurable business impact** across enterprise security, healthcare, and fintech. **MCP Foundation Development contributor**: built production MCP servers connecting LLM agents to live enterprise tools (HappyFox, Jira) for real-time context grounding, and contributed to the MCP open-source ecosystem. Proven team leader who mentored **5+ junior engineers**, led cross-functional squads of 4-8 engineers across ML, backend, and frontend, and established org-wide documentation standards. Pioneered **LLM-supervised Knowledge Distillation** (GPT-4 Teacher to distilled student ensemble) compressing 4+ hour firewall validation to 2-5 seconds. Multi-cloud architect deploying across **AWS, Azure, and Google Cloud Platform (GCP)** with **\$50K+/month** in cloud savings. Deep expertise in **multi-agent orchestration, GraphRAG, LLMOps pipelines, and regulated-domain AI** (HIPAA, SOC 2). **Zero PII leakage** across 50K+ financial documents.

WORK EXPERIENCE

Senior AI / ML Engineer

Oct 2024 - Present

Gruve AI, Bengaluru

- LLM-supervised Knowledge Distillation (Cisco ASA)**: Compressed **4+ hour firewall validation to 2-5 sec** (98% reduction) by training 3 student models via **GPT-4 Teacher** with KL-divergence loss; enabled **Fortune 500 migration** of 200+ firewalls 10x faster. Led 4 engineers; zero LLM calls at inference. [PyTorch, BERT, GPT-4, LangGraph, Neo4j, Qdrant, AWS Bedrock]
- MCP Server Development (Foundation Team)**: Built production **MCP servers** connecting **LLM agents** to HappyFox, Jira, and Confluence, eliminating **85% of manual data fetch latency**. Led 3-person squad; servers adopted by 2 internal teams. [MCP SDK, FastAPI, TypeScript, Go, WebSocket]
- TechGenie Voice Agent [MCP-Powered]**: Achieved **42% reduction in IT resolution time** (500+ weekly interactions, 80+ ticket types) by building **MCP-connected LLM agents** for real-time triage. Led 4-engineer cross-functional team. [FastAPI, Edge TTS, MongoDB, MCP, LangGraph]
- Enterprise AI Search (GSC)**: Served **10K+ queries/day at p95 <300ms** with **99.9% uptime** by building hybrid **graph + vector retrieval** (1M+ graph nodes, 50M embeddings). Mentored 2 engineers. [Neo4j, Qdrant, AWS Bedrock, ECS, LangSmith, Prometheus]
- AI Safety - PII Redaction (J.P. Morgan)**: Achieved **zero PII leakage** across 50K+ financial documents (99.2% precision, **SOC 2 passed**) by architecting **NLP redaction pipeline**; scaled to 2,000+ docs/day. [spaCy, Presidio, GLiNER, AWS, Azure]
- GUIS Meeting Intelligence Platform**: Led **5-engineer team** to build **6-service platform** (React, Express, TTS, transcription, meeting bot) for real-time meeting insights across **Google Meet** and **MS Teams**. [Node.js, React, FastAPI, Docker, Gemini, Deepgram, GCP]
- Multi-Cloud FinOps**: Saved **\$18K/month** (62% Azure compute reduction) by migrating VMs to event-driven containers across **AWS (ECS, Bedrock, SageMaker), Azure (Container Apps), and GCP (Vertex AI)**. 99.5% SLA. [Terraform, Docker, Kubernetes, Prometheus, LangSmith, MLflow]
- Team Leadership & Mentoring**: Mentored **5 junior ML engineers** (2 promoted within 8 months); established org-wide documentation standards, technical design reviews, and onboarding curriculum.

AI Engineer

Aug 2023 - Oct 2024

August AI, Bengaluru

- Multi-Agent Patient Interview System**: Raised diagnostic accuracy **67% to 91%** (5,000+ validated sessions, 2,500+ daily interactions across 3 hospitals) by building context-aware **multi-agent prompt pipeline**. Led 3-person ML team. [LangChain, GPT-4, AWS, Azure, FastAPI, LangSmith]
- Qwen 3.5 Reasoning Model Training**: Fine-tuned **Qwen 2.5 72B reasoning model** using **Unsloth** and **LLaMA Factory** on **Modal** and **RunPod GPU infrastructure** for medical domain adaptation; applied **LoRA/QLoRA** for parameter-efficient training at scale. [PyTorch, Unsloth, LLaMA Factory, Modal, RunPod, Qwen]
- HIPAA-Compliant Infrastructure**: Delivered **99.99% uptime** for 500+ concurrent users across **Azure (Container Apps, Key Vault)** and **AWS (SageMaker, ECS)**; passed 2 regulatory audits with zero findings.
- Clinical NLP Pipeline**: Achieved **\$150K annual savings** (85% manual review reduction) by building **NLP pipeline** with **97% accuracy** on 10,000+ monthly documents. Trained 2 clinical data analysts. [spaCy, PyTorch, AWS SageMaker, Azure Cognitive Services]

Machine Learning Engineer

Jan 2023 - Aug 2023

Krut AI (Snive), Bengaluru

- Generative Image Pipeline**: Reduced inference latency **8s to 1.2s (85% reduction)** (50K+ images/month) via **TensorRT**, Fast UNet, and **VAE optimizations** on **AWS G5 instances**. Mentored 2 junior engineers. [Stable Diffusion XL, TensorRT, ControlNet, AWS G5]
- GPU Cost Optimization**: Reduced GPU costs **\$32K/month** by applying **mixed precision** and gradient checkpointing; SSIM >0.95 maintained. Cut deployment time **3 days to 4 hours**. [MLflow, Docker, AWS]

Associate Data Scientist

Mar 2022 - Jan 2023

Lincode Labs Inc., Bengaluru

- Event-Driven Architecture**: Achieved **\$2M/year infrastructure savings** (450M req/month across 15 global facilities) by replacing REST polling with **Server-Sent Events** on **AWS (EC2, S3, SQS)**. [Python, REST APIs, SSE, AWS]
- CV/OCR Model Delivery**: Delivered **92% accuracy** models for 8 international clients (**0.24s inference**); reduced support tickets **60%** by training 20+ client engineers via workshops. [OpenCV, OCR, PyTorch, YOLO]

TECHNICAL SKILLS

GenAI and LLMs	GPT-4, Claude 3.5, Gemini Pro, LLaMA 3, Qwen 2.5, RAG, GraphRAG, Multi-Agent Systems, Knowledge Distillation (KL-divergence), Prompt Engineering, Fine-tuning (LoRA/QLoRA), Agentic Workflows, Reasoning Models
Orchestration and Protocols	LangChain, LangGraph, Llamaindex, Google ADK, Semantic Kernel, DSPy, NeMo Guardrails, RAGAS, MCP (Model Context Protocol), Provenance Tracking, Reranking
LLM Inference and LLMOps	vLLM, TensorRT-LLM, SGLang, LMDeploy, Ollama, Unsloth, LLaMA Factory, High-throughput Batching, Edge Deployment, LangSmith, Prometheus, MLflow, CI/CD
Cloud Platforms	AWS (SageMaker, ECS, Lambda, Bedrock, SQS, S3, G5), Azure (Container Apps, Cognitive Services, Key Vault), GCP (Vertex AI, Cloud Run, Gemini)
Vector and Data Stores	Neo4j, Qdrant, Pinecone, PostgreSQL, MongoDB, Redis, Elasticsearch, MinIO
ML / CV / Speech	PyTorch, TensorFlow, BERT, Stable Diffusion XL, YOLO, Whisper, ONNX, Quantization, spaCy, Microsoft Presidio
Infrastructure and Tools	Docker, Kubernetes, Terraform, Modal, RunPod, GitHub Actions, FastAPI, REST, gRPC
Programming Languages	Python, TypeScript, Rust, Go, JavaScript, HTML, CSS

EDUCATION

M.Sc. Big Data Analytics	St. Joseph's University, Bengaluru	2020 - 2022
B.C.A. Computer Applications	Seshadripuram College, Bengaluru	2016 - 2019

PROJECTS AND RECOGNITION

Distributed Inference Service

Multi-region ML serving at 5K req/sec with 99.9% uptime across AWS and GCP. [Ray Serve, Redis, Prometheus, Terraform]

github.com/rakshit176/distributed-inference

GSC Enterprise AI Search

Releasing publicly

1M+ graph nodes, 50M embeddings, hybrid graph + vector retrieval at p95 <300ms. [Neo4j, Qdrant, AWS Bedrock, LangSmith]

Awards: Winner - Gruve AI Hackathon (IT Help Desk) | Winner - August AI Hackathon (AI Doctor Analyzer)

Publication: Co-author, IEEE ACAI 2022 - "Water Quality Prediction using BPNN, SVR, and LSTM"

MCP Foundation Development Contributor - Open-source Model Context Protocol ecosystem